

AI POWERED RETAIL INVENTORY OPTIMIZATION SYSTEM

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: AI Powered Retail Inventory Optimization System

Technology Stack:

Python | Machine Learning | Scikit-Learn | XGBoost | Pandas | NumPy | Streamlit | Plotly
| Matplotlib | Linear Regression | ML Models

Project Description:

An AI-powered retail optimization system developed to forecast product demand, optimize inventory levels, and provide intelligent pricing recommendations. The project uses machine learning models for demand prediction, inventory risk analysis, safety stock calculation, reorder optimization, and pricing intelligence. A Streamlit dashboard is developed to visualize insights and support data-driven business decisions.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission:

18/06/2026

AI POWERED RETAIL INVENTORY OPTIMIZATION SYSTEM

1. Project Objective

The objective of this project is to develop an AI-driven retail decision support system that helps businesses optimize inventory levels, forecast product demand, and make intelligent pricing decisions.

The system uses machine learning-based demand forecasting along with inventory analytics and pricing optimization techniques to reduce stockouts, improve inventory planning, and support better business decisions.

The project includes:

- Exploratory Data Analysis
- Data preprocessing
- Feature engineering
- Demand forecasting model development
- Model comparison
- Inventory optimization
- Pricing intelligence
- Interactive Streamlit dashboard

2. Requirement Coverage

Requirement	Status	Project File / Output
Demand Forecasting Model	Completed	models/demand_forecasting2_model.pkl
Feature Engineering	Completed	notebooks/
Cleaned Dataset	Completed	data/processed/
Demand Forecast Output	Completed	data/processed/demand_forecast.csv
Inventory Optimization	Completed	data/processed/final_inventory_optimization.csv
Pricing Optimization	Completed	data/processed/pricing_optimized.csv
EDA Report	Completed	reports/figures/

Model Evaluation	Completed	notebooks/
Streamlit Dashboard	Completed	dashboard/app.py
Project Documentation	Completed	Documentation_Report.docx

3. Dataset

The project uses three retail datasets:

Demand Forecasting Dataset

Records: 10,000

Features:

- Product ID
- Date
- Store ID
- Sales Quantity
- Price
- Promotions
- Seasonality Factors
- External Factors
- Demand Trend
- Customer Segments

Inventory Dataset

Records: 10,000

Features:

- Product ID
- Store ID
- Stock Levels
- Supplier Lead Time
- Stockout Frequency
- Reorder Point
- Warehouse Capacity
- Order Fulfillment Time

Pricing Dataset

Records: 10,000

Features:

- Product ID
- Store ID
- Price
- Competitor Prices
- Discounts
- Sales Volume
- Customer Reviews
- Return Rate
- Storage Cost
- Elasticity Index

4. Work Completed

Completed an end-to-end AI/ML workflow including:

- Loaded and inspected retail datasets.
- Performed data quality validation.
- Handled missing values.
- Performed exploratory data analysis.
- Generated business-focused visualizations.
- Created demand forecasting features.
- Trained multiple machine learning regression models.
- Compared model performance.
- Selected the best performing model.
- Generated demand predictions.
- Implemented inventory risk analysis.
- Calculated safety stock.
- Generated optimized reorder points.
- Created inventory alert system.
- Performed pricing gap analysis.
- Created pricing recommendation logic.
- Developed Streamlit dashboard.
- Exported processed datasets and trained model.

5. Methodology

5.1 Data Validation

Dataset quality was analyzed by checking:

- Missing values
- Duplicate records
- Data types
- Statistical distributions

No duplicate records were identified.

5.2 Exploratory Data Analysis

Performed analysis on:

Demand Dataset:

- Sales quantity distribution
- Demand trend analysis
- Fast moving products
- Slow moving products
- Demand variability

Inventory Dataset:

- Stock level analysis
- Stockout risk
- Warehouse utilization
- Reorder analysis

Pricing Dataset:

- Price gap analysis
 - Competitor pricing comparison
 - Elasticity analysis
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5.3 Feature Engineering

Created additional analytical features:

Demand Forecasting Features

- Previous Demand
- Rolling Mean
- Product average sales

- Date based features

Inventory Features

- Warehouse Utilization
- Inventory Alert
- Safety Stock
- Recommended Reorder Point

Pricing Features

- Price Gap
- Pricing Recommendation
- Recommended Price

6. Model Development

Multiple regression models were trained and compared:

Models evaluated:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Gradient Boosting Regressor
- Extra Trees Regressor

The models were evaluated using:

- MAE
- RMSE
- R^2 Score

7. Model Results

Model	MAE	RMSE	R^2 Score
Linear Regression	125.66	145.26	-0.002
Random Forest	127.09	147.14	-0.028
XGBoost	126.83	147.70	-0.035
Extra Trees	127.59	147.77	-0.036
Gradient Boosting	125.87	145.58	-0.006

8. Selected Model

Linear Regression Model

Reason for selection:

After evaluating multiple regression algorithms, Linear Regression provided the best overall performance with the lowest error values and highest R^2 score among tested models.

Final Model:

models/demand_forecasting2_model.pkl

Performance:

Metric	Score
MAE	125.66
RMSE	145.26
R^2 Score	-0.002

9.Streamlit Application

The Streamlit application is implemented in:

dashboard/app.py

The dashboard contains:

Executive Dashboard

Displays:

- Total products
- Total stores
- Inventory alerts
- Demand overview
- Pricing actions

Demand Forecasting Page

Features:

- Actual vs predicted demand
- Forecast visualization
- Model performance metrics

- Product and store filtering

Inventory Optimization Page

Features:

- Stock monitoring
- Reorder recommendations
- Safety stock analysis
- Stockout risk identification

Pricing Optimization Page

Features:

- Competitor price comparison
- Price gap analysis
- Recommended pricing strategy

Data Explorer

Allows users to inspect processed datasets.

10. Key Findings

- Maintain optimal inventory levels using demand forecasts.
- Reorder products before stockout situations.
- Monitor products with high stockout frequency.
- Adjust pricing according to competitor prices and elasticity.
- Use forecasting insights for future planning.

10. Recommendations

- Monitor low-rated products.
- investigate recurring product issues.
- prioritize quality checks for high-risk orders.

11. Achievements

Successfully completed:

- Data preprocessing
- Exploratory analysis

- Machine learning model development
- Model evaluation
- Demand forecasting
- Inventory optimization
- Pricing optimization
- Dashboard deployment

12. Limitations

- Dataset contains limited external business factors.
- Demand prediction performance is limited by available historical information.
- Real-time sales integration is not implemented.
- Pricing recommendations are rule-based rather than dynamic reinforcement learning.

13. Final Submission Contents

- Documentation Report
- Machine Learning Model
- Streamlit Dashboard
- Processed Datasets
- EDA Visualizations
- Jupyter Notebooks
- README.md
- requirements.txt

14. How to Run

```
pip install -r requirements.txt  
streamlit run dashboard/app.py
```

15. Final Conclusion

The AI Powered Retail Inventory Optimization System successfully delivers an end-to-end machine learning solution for demand forecasting, inventory monitoring, and pricing optimization.

The system combines machine learning predictions with business analytics to help retailers reduce stockouts, improve inventory efficiency, and make data-driven decisions through an interactive dashboard.